APPLIED AI & INSIGHTS LAB

MSc Data Science and Artificial Intelligence

Middlesex University · Faculty of Science and Technology

When the Picture Fails, Do the Words Take Over?

Multimodal Lost-Person Search Using Image and Text Queries Across Multiple Cameras

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POSTER NO. 7



OUR KEY FINDINGS

A written description is not a weaker copy of a photograph. Against a conventional re-ID photo model it becomes **the better evidence once the person is small** — and that point can be measured.

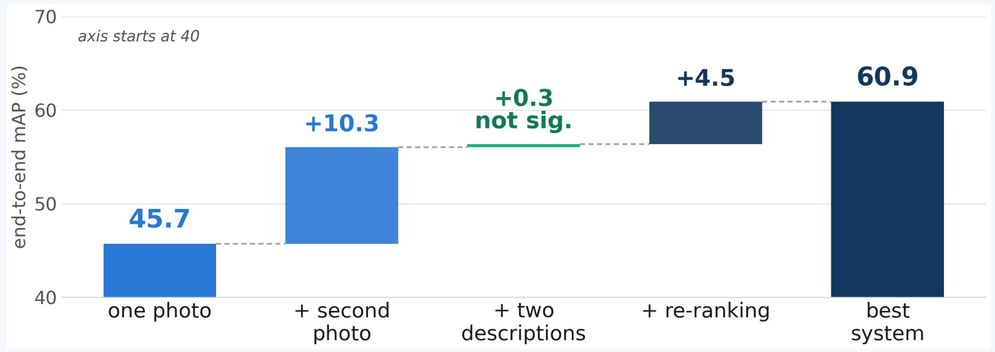
32 px
from this height down, the description beats the appearance-model photo

+7.6 to 12.3
mAP points a description adds to the appearance model, at every resolution tested

≤ 0.5
points it adds to a strong vision-language encoder

Tested on **1,016 searches · 217 held-out people** · 95% person-level intervals · nothing trained

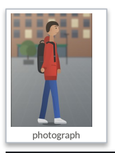
Best system: where its +15.2 points over one photo come from



A second photo and re-ranking do the work. With a strong vision-language encoder the description adds almost nothing in late fusion.

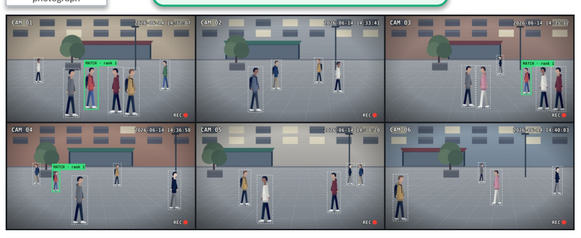
INTRODUCTION

A witness can give two clues: **a photograph** and **a description**. Camera networks record everything, but searching them is still done by hand.


photograph

+ description from a witness: **"man, red jacket, blue jeans, black backpack"**

searched across **6 cameras · 11,816 frames**



353,000
UK missing-person reports a year

1 billion+
surveillance cameras worldwide

≈ 23 px
a face at median height: too small to identify

WHAT THIS PROJECT ADDS

In the 41 studies reviewed	This project
Image OR text query; the few that take both are tested on clean images	**Photo + description on whole frames, at 7 resolution levels**
No measure of how a description's value changes as resolution falls	**Measured: at 16 px text keeps 31% of its accuracy, the appearance model 12%**
One fixed fusion weight, whatever the image quality	**Weight fitted per encoder and per resolution; it shifts toward text**
One description per search (PS-ReID's set: GPT-4o drafts, then edited)	**Two independent human descriptions; the writer moves accuracy 16.7 points**
Small people filtered out (PS-ReID's set drops < 75 px)	**Small people kept; recall reported by person size**

Research questions → answers

1. **Can people be found across cameras from whole, uncropped frames?**
   Yes, but detection sets the ceiling: only 50.2% of people 64–96 px tall reach the gallery.
2. **How should a photograph and a description be combined?**
   Standardise both scores and weight them with α fitted per encoder; α moves toward text as images worsen.
3. **When does the description beat the photograph, and by how much?**
   From 32 px down against the appearance model (by 2.1 to 5.1 points); never against the vision-language model.

METHODS

1. **Detect** — YOLO11m finds every person in each frame
2. **Encode** — OSNet: people → vectors; IRRA (CLIP): people + text
3. **Match** — score every candidate against the query
4. **Fuse** — weigh the photo against the description
5. **Re-rank** — k-reciprocal, adapted to photo + text
6. **Sightings** — cameras that saw the person, in order (demo app)

■ red steps: where this project makes its own contribution

$$\text{fused score} = \alpha \cdot \text{photo} + (1 - \alpha) \cdot \text{description}$$

both scores standardised per query · α fitted on validation people, separately for each encoder and each resolution

Data

PRW
11,816 frames
6 cameras · 932 people

PRW-TBPS
2 independent human descriptions per search

Market-1501
checks the photo model loaded correctly

Distance is simulated: gallery crops are shrunk to 96, 64, 48, 32, 24 and 16 px and resized back, so only the pixels change. The query photo and description stay sharp.

How we measured

End-to-end mAP
people the detector missed count as failures

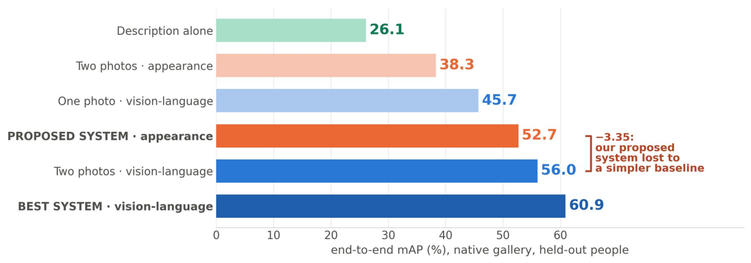
Separate people
settings fitted on 225 people, results on others

Person-level bootstrap
5,000 resamples, paired 95% intervals

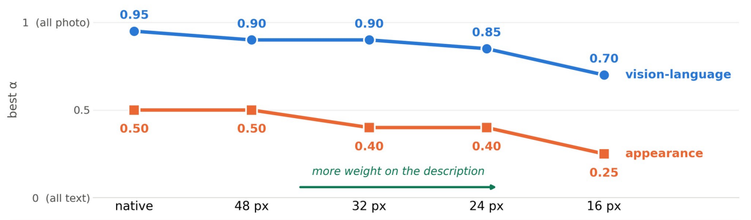
Nothing was trained. Published weights only, so only the method, the encoder and the image quality vary. The evaluation was audited, 4 flaws fixed and every result re-run.

RESULTS

Which system wins?



The best balance shifts toward the words



Quality-aware weighting (the original idea): choosing α per resolution condition beats a fixed α tuned on normal footage: +1.86 points at 16 px with the appearance model. Weighting each candidate by its own size gave no gain for that model.

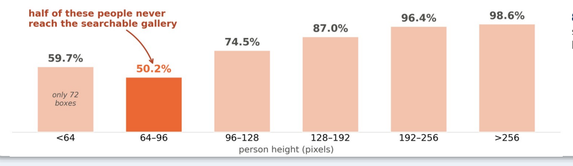
Same man, two writers, opposite results

Writer 1 "The bald man walking on the street is wearing a white shirt and a pair of black pants." — **0.5%**

Writer 2 "The fat, dark skinned, bald man, with a pink short sleeve shirt on his upper body and a pair of black trousers, is walking on the sidewalk..." — **100%**

Real PRW-TBPS descriptions of one search; boxes show that search's AP with the description alone. On average the writer moves accuracy by 16.7 points.

Detection sets a ceiling



86.2% of all people reach the searchable gallery; small people are lost before matching starts

CONCLUSIONS

32 px
Words beat pixels
From 32 px down, a description beats the appearance-model photograph.

3×
Words rescue tiny people
At 16 px, adding a description triples the appearance model's accuracy (3.7 → 11.3).

+15.2
Invest in the image first
Best system over one photo: a second photo (+10.3) and re-ranking (+4.5) do the work.

+7.4
Ask two witnesses
Averaging two descriptions adds 7.4 points; which writer you get moves accuracy 16.7 points.

+1.86
Weight words by image quality
One α per resolution beats a fixed α tuned on normal footage: +1.86 points at 16 px (appearance model).

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References

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2. Jiang & Ye (2023) Cross-modal implicit relation reasoning and aligning for text-to-image person retrieval (IRRA), CVPR.
3. Zhou et al. (2019) Omni-scale feature learning for person re-ID (OSNet), ICCV.
4. Zhong et al. (2017) Re-ranking person re-ID with k-reciprocal encoding, CVPR.
5. Zhang et al. (2023) Text-based person search in full images (PRW-TBPS), ACM MM Workshop.
6. Yan et al. (2025) PS-ReID, arXiv:2503.21595.

Icons: Material Design Icons (Pictogrammers). CCTV scenes and figures are illustrations.